

JEE Main 2025
Mock Practice Test -1

Time Allowed: 3 Hours

Maximum Marks : 300

PART-A: MATHEMATICS

SECTION-I

1. Which of the following is not empty set

- (A) $A = \{x : 1 < x < 2 \text{ and } x \in \mathbb{N}\}$
(B) $B = \{x : x^2 - 2 = 0 \text{ and } x \text{ is rational}\}$
(C) $C = \{x : x \text{ is even prime number } > 2\}$
(D) $D = \{x : x^2 = 0 \text{ and } x \text{ is integer}\}$

Ans. (D)

Sol. $A = \phi$ because there are no natural numbers between 1 and 2.

$B = \phi$ because $x^2 = 2 \Rightarrow x = \pm \sqrt{2}$ not a rational number

$C = \phi$ because there is only one even prime 2

2. If the set of factors of a whole number 'n' including 'n' itself but not '1' is denoted by $F(n)$. If

$F(16) \cap F(40) = F(x)$ then 'x' is

- (A) 4
(B) 8
(C) 6
(D) 10

Ans. (B)

Sol. $F(16) = \{2, 4, 8, 16\}, F(40) = \{2, 4, 5, 8, 10\}$

$F(16) \cap F(40) = \{2, 4, 8\} = F(8)$

$F(x) = F(8) \Rightarrow x = 8$

3. Which of the following is finite set

- (A) $A = \{x : x \text{ is set of points on a line}\}$
(B) $B = \{x : x \in \mathbb{N} \text{ and } x \text{ is prime}\}$
(C) $C = \{x : x \in \mathbb{N} \text{ and } x \text{ is odd}\}$
(D) $D = \{x : x \in \mathbb{N} \text{ and } (x-1)(x-2) = 0\}$

Ans. (D)

Sol. $x-1=0, x-2=0 \Rightarrow x=1, 2$

4. Which of the following pairs of sets are equal

- (A) $A = \{x : x \text{ is letter of word "ALLOY"}\}, B = \{x : x \text{ is letter of word "LOYAL"}\}$
(B) $A = \{-2, -1, 0, 1, 2\}, B = \{1, 2\}$
(C) $A = \{0\}, B = \{x : x > 5 \text{ and } x < 15\}$
(D) $A = \{x : x-5 = 0\}, B = \{x : x^2 = 25\}$

Ans. (A)

Sol. $A = \{a, l, o, y\}, B = \{a, l, o, y\}$

5. Let $A = \{1, \{2, 3\}\}$ then the number of subsets of A are

- (A) 2
(B) 4

(C) 8

(D) 0

Ans. (B)

Sol. $n(A) = 2$

no of subsets $= 2^n = 2^2 = 4$

6. How many elements has $P(A)$, if $A = \phi$

(A) 1

(B) 2

(C) 3

(D) 0

Ans. (A)

Sol. $n(A) = 0$, $n[P(A)] = 2^n = 2^0 = 1$

7. In a group of 69 people, 37 like coffee, 52 like tea and each person like at least one of the two drinks. The number of persons liking both coffee and tea is

(A) 16

(B) 13

(C) 19

(D) 20

Ans. (D)

Sol. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

We have, $69 = 37 + 52 - n(A \cap B)$

$n(A \cap B) = 89 - 69 = 20$

8. $|x - 3| + 2|x + 1| = 4$, then $x =$

(A) -1

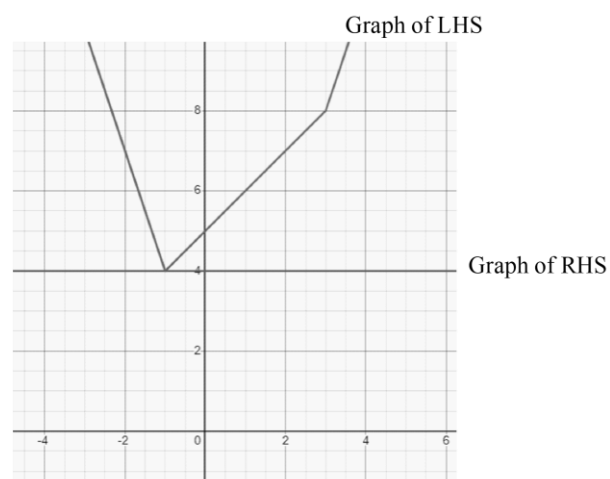
(B) 1

(C) 0

(D) None

Ans. (A)

Sol.

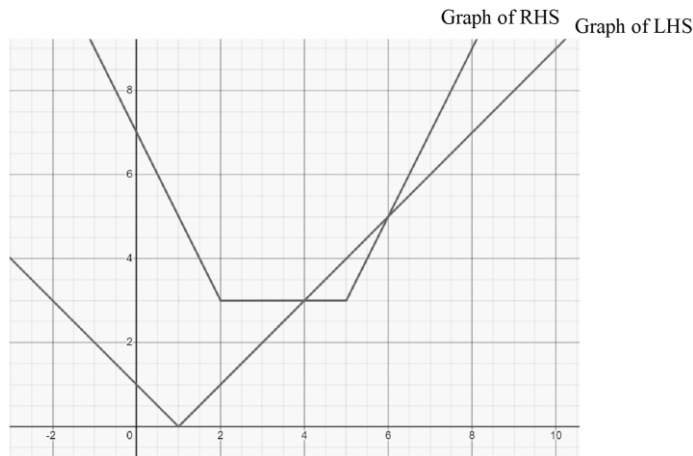


Point of intersection of the two graphs is the solution. $x = -1$

9. Number of real x satisfying the equation $|x - 1| = |x - 2| + |x - 5|$ is
- (A) 1
(B) 2
(C) 3
(D) more than 3

Ans. (B)

Sol.



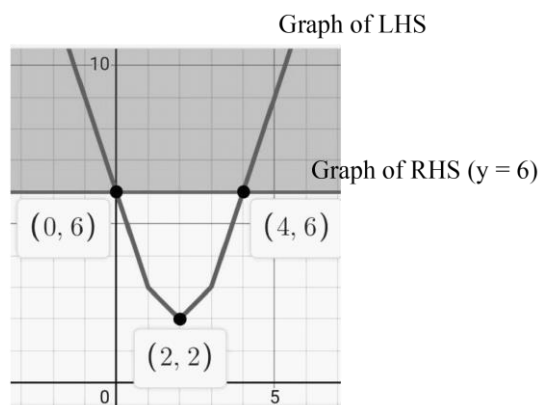
$\therefore$ Point of intersections = 2

$\therefore$ No. of solutions = 2

10. If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then
- (A) $0 \leq x \leq 4$
(B) $x \leq -2$ or $x \geq 4$
(C) $x \leq 0$ or $x \geq 4$
(D) none of these

Ans. (C)

Sol.



Final region will be where $LHS \geq RHS$ so check values of ' x '

11. For the equation $\log_{100} |x + y| = \frac{1}{2}$ and $\log_{10} y - \log_{10} |x| = \log_{100} 4$, (x, y) is
- (A) $(10, 20/3)$
(B) $(10, 20)$
(C) $(-10, 20)$
(D) $(-10/3, -20/3)$

Ans. (C)

Sol. $\log_{100} |x + y| = \frac{1}{2}, \log_{10} y - \log_{10} |x| = \log_{100} 4 = \log_{10} 2$

$$\Rightarrow |x + y| = 10 \text{ and } \frac{y}{|x|} = 2$$

$$\Rightarrow x + y = \pm 10 \text{ and } y = 2|x|$$

$$\text{From these two, } x + 2|x| = \pm 10$$

$$\text{Now when } x > 0, x + 2x = \pm 10 \Rightarrow x = 10/3 \Rightarrow y = 20/3$$

$$\text{and when } x < 0, x - 2x = \pm 10 \Rightarrow x = -10 \Rightarrow y = 20$$

Hence, the solutions are $(-10, 20)$ and $(10/3, 20/3)$

- 12.** The value of the expression $\log_{10}(\tan 6^\circ) + \log_{10}(\tan 12^\circ) + \log_{10}(\tan 18^\circ) + \dots + \log_{10}(\tan 84^\circ)$ is
- (A) a whole number
 (B) an irrational number
 (C) a negative integer
 (D) a rational number which is not an integer

Ans. (A)

Sol. $E = \log_{10}(\tan 6^\circ) + \log_{10}(\tan 12^\circ) + \log_{10}(\tan 18^\circ) + \dots + \log_{10}(\tan 84^\circ)$

$$= \log_{10}((\tan 6^\circ \times \tan 84^\circ)(\tan 12^\circ \times \tan 78^\circ) \dots (\tan 42^\circ \times \tan 48^\circ))$$

$$= \log_{10}((\tan 6^\circ \times \cot 6^\circ)(\tan 12^\circ \times \cot 12^\circ) \dots (\tan 42^\circ \times \cot 42^\circ))$$

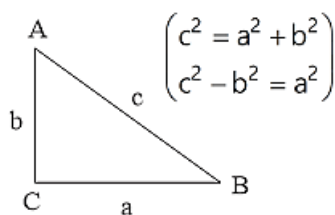
$$= \log_{10} 1 = 0$$

- 13.** Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a} (b+c \neq 1, c-b \neq 1)$ equals
- (A) 1
 (B) 2
 (C) 3
 (D) $\frac{1}{2}$

Ans. (B)

Sol. $N^r = \frac{1}{\log(b+c)} + \frac{1}{\log(c-b)} = \frac{\log_a(c^2 - b^2)}{\log(b+c) \cdot \log_a(c-b)}$

$$D^r = \frac{1}{\log_a(b+c)} \times \frac{1}{\log_a(c-b)}$$



$$\frac{N^r}{D^r} = \log_a(c^2 - b^2) = \log_a a^2 = 2$$

14. If $\log_{10} 2 = 0.3010$ & $\log_{10} 3 = 0.4771$, find the value of $\log_{10} (2.25)$.

- (A) 0.3522
- (B) 0.03522
- (C) 1.3522
- (D) None

Ans. (A)

Sol. $\log_{10} 2.25$

$$= \log_{10} 9/4 = 2\log_{10} 3/2$$

$$= 2(\log 3 - \log 2) = 2(0.4771 - 0.3060) = 2(0.1761) = 0.3522$$

15. Let $A = \{x \mid x \text{ is a prime number and } x < 30\}$. The number of different rational numbers whose numerator and denominator belong to A is

- (A) 90
- (B) 180
- (C) 91
- (D) None

Ans. (C)

Sol. $A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$. A rational number is made by taking any two in any order. So, the required number of rational numbers is (including 1)

$${}^{10}P_2 + 1 \text{ (for rational no = 1) and } {}^nP_r = \frac{n!}{(n-r)!} = 10!/8! + 1 = 90 + 1 = 91$$

16. If $xy^2 = 4$ and $\log_3 (\log_2 x) + \log_{1/3} (\log_{1/2} y) = 1$, then x equals

- (A) 4
- (B) 8
- (C) 16
- (D) 64

Ans. (D)

Sol. $\log_3 (\log_2 x) + \log_{1/3} (\log_{1/2} y) = 1$

$$\text{or } \log_3 (\log_2 x) - \log_3 (\log_{1/2} y) = 1$$

$$\text{or } \log_3 (\log_2 (4/y^2)) - \log_3 (\log_{1/2} y) = 1$$

$$\log_3 \left(\frac{\log_2 \frac{4}{y^2}}{\log_{\frac{1}{2}} y} \right) = 1$$

$$\frac{\log_2 \frac{4}{y^2}}{-\log_2 y} = 3; \log_y \frac{4}{y^2} = -3$$

$$y^{-3} = \frac{4}{y^2} \Rightarrow y = \frac{1}{4} \Rightarrow x = 64$$

17. If $x = -5 + 2\sqrt{-4}$, then $x^4 + 9x^3 + 35x^2 - x + 4 = ?$

- (A) 160
- (B) - 160
- (C) 41
- (D) 0

Ans. (B)

Sol. $x^2 + 9x^3 + 35x^2 - x + 4 \quad \because x^2 + 10x + 41 = 0$

$$x^2(x^2 + 10x + 41) - x^3 - 6x^2 - x + 4$$

$$x^2(0) - x(x^2 + 10x + 41) + 4x^2 + 40x + 4$$

$$x^2(0) - x(0) + 4(x^2 + 10x + 41) - 160$$

$$x^2(0) - x(0) + 4(0) - 160 = (-160)$$

18. The equation $\left(\sqrt{1 + \log_x \sqrt{27}}\right) \log_3 x + 1 = 0$ has

- (A) one integral solution
- (B) two irrational solution
- (C) two real solution
- (D) one prime solution

Ans. (C)

Sol. $\left(\sqrt{1 + \log_x \sqrt{27}}\right) \log_3 x + 1 = 0$

$$\left(\sqrt{1 + \log_x 3^{3/2}}\right) \log_3 x + 1 = 0$$

$$\left(\sqrt{1 + \frac{3}{2} \log_x 3}\right) \log_3 x + 1 = 0$$

$$\text{Let } \log_x 3 = t$$

$$\left(\sqrt{1 + \frac{3}{2}t}\right) \frac{1}{t} + 1 = 0$$

$$\sqrt{1 + \frac{3}{2}t} = -t$$

$$1 + \frac{3t}{2} = t^2$$

$$2t^2 - 3t - 2 = 0$$

$$2t^2 - 4t + t - 2 = 0$$

$$2t(t - 2) + 1(t - 2) = 0$$

$$t = -\frac{1}{2}, 2$$

$$\log_x 3 = 2$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

$$x = \sqrt{3} \text{ as } x \text{ is positive}$$

$$\log_x 3 = -\frac{1}{2}$$

$$x = 3^{-2} \Rightarrow x = \frac{1}{9}$$

19. If $n(A) = 20$, $n(B) = 30$, $n(A \cup B) = 47 = a$, $n(A \cap B) = b$, then a and b are

- (A) co prime
- (B) twin prime
- (C) a is multiple of b
- (D) $\text{hcf}(a, b) = 10$

Ans. (A)

$$\text{Sol. } n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= 20 + 30 - 47 \Rightarrow b = 3$$

a, b are co prime

20. If $4^A + 9^B = 5^C$ where $A = \log_{16} 4$, $B = \log_3 9$, $C = \log_x 83$. If $\log_3 \log_7 y = 0$, then x and y are

- (A) twin prime
- (B) complex numbers
- (C) irrational numbers
- (D) $\text{hcf}(x, y) = 2$

Ans. (A)

$$\text{Sol. } A = \frac{1}{2}, B = 2$$

$$4^{1/2} + 9^2 = 5^C$$

$$2 + 81 = 5^{\log_x 83}$$

$$83 = 5^{\log_x 83} \Rightarrow 83^{\log_x 5} = 83$$

$$\log_x^5 = 1 (x = 5)$$

$$\log_3 \log_7 y = 0$$

$$3^0 = \log_7 y \Rightarrow 1 = \log_7 y \Rightarrow y = 7$$

x, y are twin prime

SECTION-II

21. The value of $a^3 + b^3 + c^3 - 3abc$ if $a + b + c = 5$ and $a^2 + b^2 + c^2 = 29$

Ans. 155

$$\text{Sol. } (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$25 = 29 + 2(ab + bc + ca)$$

$$ab + bc + ca = -2$$

$$a^2 + b^2 + c^2 - 3abc = (a + b + c)((a^2 + b^2 + c^2) - (ab + bc + ca))$$

$$= (5)(29 + 2) = 5(31) = 155$$

22. $\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc}$ has the value equal to (given $a, b, c > 0$)

Ans. 1

Sol. $\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc}$

$$= \log_{abc} \sqrt{bc} + \log_{abc} \sqrt{ac} + \log_{abc} \sqrt{ab}$$

$$= \log_{abc} \sqrt{ab} \times \sqrt{bc} \times \sqrt{ca}$$

$$= \log_{abc} abc = 1$$

23. The value of the expression $\frac{2}{\log_4 (2000)^6} + \frac{3}{\log_5 (2000)^6}$ is equal to $1/k$ then $k =$

Ans. 6

Sol. $\frac{2}{\log_4 (2000)^6} + \frac{3}{\log_5 (2000)^6}$

$$= 2 \log_{(2000)^6} 4 + 3 \log_{(2000)^6} 5$$

$$= \frac{2}{6} \log_{2000} 4 + \frac{3}{6} \log_{2000} 5$$

$$= \frac{1}{6} \log_{2000} 4^2 + \frac{1}{6} \log_{2000} 5^3$$

$$= \frac{1}{6} (\log_{2000} 4^2 + \log_{2000} 5^3)$$

$$= \frac{1}{6} (\log_{2000} 4^2 \times 5^3)$$

$$= \frac{1}{6} (\log_{2000} 2000) = \frac{1}{6}$$

24. Let 'm' be the number of digits in 3^{40} and 'p' be the number of zeroes in 3^{-40} after decimal before starting a significant digit then $(m + p)$ is ($\log 3 = 0.4771$)

Ans. 39

Sol. Given $\log_{10} 3 = 0.4771$

$$\log_{10} 3^{40} = 40 \times \log_{10} 3$$

$$= 40 \times 0.4771 = 19.084$$

no. of digits = 20

$$\log_{10} 3^{-40} = -40 \times \log_{10} 3 = -40 \times 0.4771$$

$$= -19.084$$

$\therefore$ 19 0's after decimal

25. If x satisfies the inequality $\log_{(x+3)}(x^2 - x) < 1$, then number of integral values of x

Ans. 0

Sol. $\log_{(x+3)}(x^2 - x) < 1$

For the inequality to be defined,

$$x^2 - x > 0 \text{ and } x + 3 > 0$$

$$\Rightarrow x(x - 1) > 0 \text{ and } x > -3$$

$$\Rightarrow x \in (-\infty, 0) \cup (1, \infty) \text{ and } x \in (-3, \infty)$$

$$\therefore x \in (-3, 0) \cup (1, \infty)$$

Case 1: When $x + 3 > 1$ i.e., $x \in (-2, 0) \cup (1, \infty)$

$$x^2 - x < x + 3 \Rightarrow x^2 - 2x - 3 < 0$$

$$\Rightarrow (x - 3)(x + 1) < 0$$

$$\Rightarrow x \in (-1, 3)$$

$$\therefore x \in (-1, 0) \cup (1, 3) \dots(1)$$

Case 2: When $0 < x + 3 < 1$ i.e., $x \in (-3, -2)$

$$x^2 - x > x + 3$$

$$\Rightarrow x^2 - 2x - 3 > 0 \Rightarrow (x - 3)(x + 1) > 0$$

$$\Rightarrow x \in (-\infty, -1) \cup (3, \infty)$$

$$\therefore x \in (-3, -2) \dots(2)$$

Hence, from (1) and (2),

$$X \in (-3, -2) \cup (-1, 0) \cup (1, 3)$$

26. Of the members of three athletic teams in a school 21 are in the cricket team, 26 are in the hockey team and 29 are in the football team. Among them, 14 play hockey and cricket, 15 play hockey and foot ball, and 12 play foot ball and cricket. Eight play all the three games. The total number of members in the three athletic teams is

Ans. 43

Sol. $n(C) = 21, n(H) = 26, n(F) = 29,$

$$n(H \cap C) = 14, n(H \cap F) = 15$$

$$n(F \cap C) = 12, n(F \cap C \cap H) = 8$$

$$\text{Total number of players} = n(C \cup H \cup F) = 43$$

$$n(C \cup H \cup F) = n(C) + n(H) + n(F) - n(C \cap H) - n(H \cap F) - n(F \cap C) + n(C \cap H \cap F)$$

$$= 21 + 26 + 29 - 14 - 15 - 12 + 8 = 43$$

27. In a town of 10,000 families it was found that 40% families buy news paper A, 20% families buy news paper B, and 10% families buy news paper C. Also 5% families buy A and B, 3% buy B and C, 4% buy A and C, and 2% buy all the three news papers. Then the number of families which buy exactly one of A, B, C is

Ans. 5200

Sol. Number of families which buy exactly one of A, B, C

$$= n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(C \cap B) - 2n(A \cap C) + 3n(A \cap C \cap B)$$

- 28.** Suppose $A_1, A_2, \dots, A_{30}$ are thirty sets each with five elements and $B_1, B_2, \dots, B_n$ are n sets each with three elements such that $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$. If each element of S belongs to exactly ten of the A_i 's and exactly 9 of the B_j 's, then the value of n is

Ans. 45

Sol. $S = \bigcup_{i=1}^{30} A_i \Rightarrow n(S) = \frac{1}{10} (5 \times 30) = 15$

Again, $S = \bigcup_{j=1}^n B_j \Rightarrow n(S) = \frac{1}{9} (3 \times n) = \frac{n}{3}$

Thus $\frac{n}{3} = 15 \Rightarrow n = 45$

- 29.** In a class of 55 students the numbers of students studying different subjects are 23 in mathematics, 24 in physics, 19 in chemistry, 12 in mathematics and physics, 9 in mathematics and chemistry, 7 in physics and chemistry and 4 in all the three subjects. the numbers of students who have taken exactly one subject is

Ans. 22

Sol. Let M, P, C denote the set of students studying Mathematics, Physics and Chemistry respectively.

$n(M) = 23, n(P) = 24, n(C) = 19, n(M \cap P) = 12, n(M \cap C) = 9, n(P \cap C) = 7, n(M \cap P \cap C) = 4$

Now, $n(M \cap P' \cap C')$

$n(M \cap (P \cup C)')$

$n(M) - n(M \cap (P \cup C))$

$= n(M) - n[(M \cap P) \cup (M \cap C)]$

$= n(M) - [n(M \cap P) + n(M \cap C)] + n(M \cap P \cap C) = 23 - [12 + 9] + 4 = 6$

Similarly, $n(P \cap M' \cap C')$

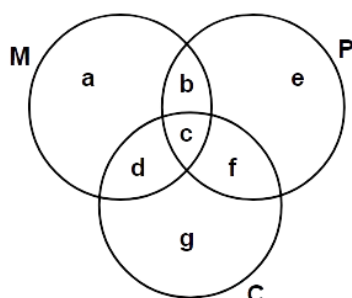
$= n(P) - [n(P \cap M) + n(P \cap C)] + n(M \cap P \cap C) = 24 - [12 + 7] + 4 = 9$

and $n(C \cap P' \cap M')$

$= n(C) - [n(C \cap P) + n(C \cap M)] + n(M \cap P \cap C) = 19 - [9 + 7] + 4 = 7$

Hence, number of students studying exactly one subject $= 6 + 9 + 7 = 22$

Alternatively, using Venn diagram



Given, $a + b + c + d = 23 \dots (1)$

$b + c + e + f = 24 \dots (2)$

$d + c + f + g = 19 \dots (3)$

$b + c = 12, d + c = 9, c + f = 7$ and $c = 4$

$\therefore b = 8, d = 5, f = 3$

From (1), $a = 6$

From (2), $e = 9$

From (3), $g = 7$

Required number = $a + e + g = 22$

30. $a > 0 (a \neq 1), b > 0 (b \neq 1)$ such that $a^{(\log_b^b)^x} = b^{(\log_a^a)^x}$ then $2x =$

Ans. 1

Sol. Taking $\log_b$ both sides we get

$$(\log_a^b)^x \log_b^a = (\log_b^a)^x$$

$$\therefore (\log_a^b)^x = (\log_b^a)^{x-1}$$

$$\therefore 1 - x = x \Rightarrow x = \frac{1}{2}$$

PART-B: PHYSICS

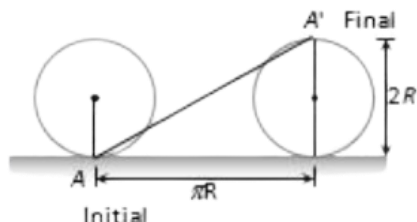
SECTION-I

31. A wheel of radius 1 m rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is

- (A) 2π
- (B) $\sqrt{2}\pi$
- (C) $\sqrt{\pi^2 + 4}$
- (D) π

Ans. (C)

Sol. Horizontal distance covered by the wheel in half revolution = πR .



32. A particle travelling along a straight-line traverse one third of the total distance with a velocity V_0 . The remaining part of the distance was covered with a velocity V_1 for half the time and velocity V_2 for the other half of the time. Find the mean velocity of the particle averaged over the whole motion of time:

- (A) $\frac{3V_0(V_1 + V_2)}{V_1 + V_2 + 4V_0}$
- (B) $\frac{V_0(V_1 + V_2)}{V_1 + V_2 + 12V_0}$
- (C) $\frac{V_0}{V_1 + V_2 + V_0}$
- (D) $\frac{V_1 + V_2 + V_3}{V_0}$

Ans. (A)

Sol. Let the time taken for one third distance be t_1 , then $t_1 = \frac{d}{3V_0}$, where d is the total length of the journey. Let the time taken for next $2d/3$ distance be t_2 ,

$$\frac{2d}{3} = \frac{V_1 t_2}{2} + \frac{V_2 t_2}{2} = \frac{(V_1 + V_2) t_2}{2}$$

$$t_2 = \frac{4d}{3(V_1 + V_2)}. \text{ Thus, the total time taken for the journey is } \frac{d}{3V_0} + \frac{4d}{3(V_1 + V_2)} = \frac{d(V_1 + V_2 + 4V_0)}{3V_0(V_1 + V_2)}$$

Thus the average velocity = $\frac{\text{Total distance}}{\text{Total time}}$

$$V = \frac{d}{\frac{d(V_1 + V_2 + 4V_0)}{3V_0(V_1 + V_2)}} = \frac{3V_0(V_1 + V_2)}{(V_1 + V_2 + 4V_0)}$$

33. Velocity of a body moving along a straight line with uniform retardation reduces by $3/4^{\text{th}}$ of its initial velocity in time t_0 . The total time of motion of the body till its velocity becomes zero is

(A) $\frac{4}{3}t_0$

(B) $\frac{3}{2}t_0$

(C) $\frac{5}{3}t_0$

(D) $\frac{8}{3}t_0$

Ans. (A)

Sol. $\frac{u}{4} = u - at_0$

$$\therefore a = \frac{3u}{4t_0}$$

$$\text{Or } \frac{u}{a} = \frac{4}{3}t_0$$

$$\text{Now } 0 = u - at \text{ or } t = \frac{u}{a} = \frac{4}{3}t_0$$

34. A body starts from rest with uniform acceleration a , its velocity after n seconds is v . The displacement of the body in last 3s is (assume total time of journey from 0 to n second)

(A) $\frac{v(6n-9)}{2n}$

(B) $\frac{2v(6n-9)}{n}$

(C) $\frac{2v(2n+1)}{n}$

(D) $\frac{2v(n-1)}{n}$

Ans. (A)

Sol. $v = an$

$$\therefore a = \left(\frac{v}{n} \right)$$

$$\text{Now, } S_n = \frac{1}{2}an^2 = \frac{1}{2}\left(\frac{v}{n}\right)(n^2) = \frac{vn}{2}$$

$$\text{And } S_{n-2} = \frac{1}{2}a(n-2)^2$$

$$= \frac{1}{2}\left(\frac{v}{n}\right)(n-2)^2$$

$\therefore$ Displacement in last 2 s will be

$$S = S_n - S_{n-2}$$

$$= \frac{v}{2} \left[n - \frac{(n-2)^2}{n} \right]$$

$$\left(\frac{6n-4}{n} \right) \frac{v}{2}$$

- 35.** Two trains, each of length 100m, moving in opposite directions along parallel lines, meet each other with speeds of 50 kmh^{-1} and 40 kmh^{-1} . If their accelerations are 30 cms^{-2} and 20 cms^{-2} , respectively, find the time they will take to pass each other.

(A) $10\sqrt{40} - 40 \text{ s}$

(B) $10\sqrt{33} - 50 \text{ s}$

(C) $10\sqrt{22} - 50 \text{ s}$

(D) $10\sqrt{42} - 40 \text{ s}$

Ans. (B)

Sol. $V_{\text{rel}} = 50 + 40 = 90 \text{ kmh}^{-1} = 25 \text{ ms}^{-1}$

$$a_{\text{rel}} = 30 + 20 = 50 \text{ cms}^{-2} = 0.5 \text{ ms}^{-2}$$

$$s_{\text{rel}} = u_{\text{rel}}t + \frac{1}{2}a_{\text{rel}}t^2$$

$$\Rightarrow 100 + 100 = 25t + \frac{1}{2}0.5t^2$$

$$\text{Solving, we get } t = 10\sqrt{33} - 50 \text{ s}$$

- 36.** A ball is dropped from the top of a tower, it takes "t" seconds to cover lower half of the tower, then its total time of fall is

- (A) $\frac{t}{\sqrt{2}}$
 (B) $\frac{t}{\sqrt{2} - 1}$
 (C) $\frac{\sqrt{2} t}{\sqrt{2} - 1}$
 (D) $\frac{t\sqrt{2}}{\sqrt{2} + 1}$

Ans. (C)

Sol. $T = \sqrt{\frac{2h}{g}}$

$$t = \sqrt{\frac{2h}{g}} - \sqrt{\frac{2\left(\frac{h}{2}\right)}{g}} = \sqrt{\frac{2h}{g}} \left(1 - \frac{1}{\sqrt{2}}\right) = T \left(\frac{\sqrt{2} - 1}{\sqrt{2}}\right)$$

$$\therefore T = \frac{\sqrt{2} t}{\sqrt{2} - 1}$$

- 37.** A stone is dropped from the top of a tower of height h. After 1 second, another stone is dropped from the balcony 20 m below the top, both reach the bottom simultaneously. What is the value of h?

- (A) 31.25 m
 (B) 100 m
 (C) 120 m
 (D) 130 m

Ans. (A)

Sol. According to the equation

$$h = ut + \frac{1}{2}gt^2$$

$$h = \frac{1}{2}gt^2 \dots\dots(1)$$

$$(h - 20) = 0 + \frac{1}{2}g(t - 1)^2$$

$$(h - 20) = \frac{1}{2}g(t - 1)^2 \dots(2)$$

$u = 0$ in both case because stones are dropped from rest.

From the equation (1) and (2) we get

$$h = \frac{1}{2}gt^2$$

$$(h - 20) = \frac{1}{2}g(t - 1)^2$$

$$h - (h - 20) = \frac{1}{2}gt^2 - \frac{1}{2}g(t - 1)^2$$

$$h - h + 20 = \frac{1}{2}gt^2 - \frac{1}{2}gt^2 - \frac{1}{2}g + gt$$

$$20 = gt - \frac{g}{2}$$

$$\Rightarrow gt = 25 \quad (\because g = 10 \text{ ms}^{-2})$$

$$\Rightarrow t = 2.5 \text{ second}$$

$$h = \frac{1}{2} \times 10 \times (2.5)^2 = 31.25 \text{ m}$$

- 38.** A balloon rises from rest on the ground with constant acceleration $g/8$. A stone is dropped from the balloon when the balloon has risen to a height of H . Find the time taken by the stone to reach the ground.

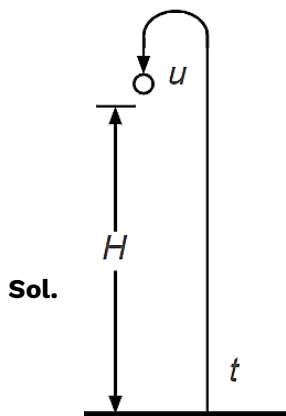
(A) $\sqrt{\frac{H}{g}}$

(B) $2\sqrt{\frac{H}{g}}$

(C) $2\sqrt{\frac{g}{H}}$

(D) $4\sqrt{\frac{g}{H}}$

Ans. (B)



$$u = \sqrt{2 \times \frac{g}{8} H} \Rightarrow u = \frac{\sqrt{gH}}{2}$$

$$-H = ut - \frac{1}{2}gt^2$$

$$\Rightarrow gt^2 - 2ut - 2H = 0$$

$$gt^2 - \sqrt{gH}t - 2H = 0$$

$$\Rightarrow gt^2 - 2\sqrt{gH}t + \sqrt{gH}t - 2H = 0$$

$$\sqrt{gt}[\sqrt{gt} - 2\sqrt{H}] + \sqrt{H} = [\sqrt{gt} - 2\sqrt{H}] = 0$$

$$\Rightarrow (\sqrt{gt} + \sqrt{H})(\sqrt{gt} - 2\sqrt{H}) = 0 \Rightarrow t = 2\sqrt{\frac{H}{g}}$$

- 39.** A train is moving at a constant speed 'V' when its driver observes another train in front of him on the same track and moving in same direction with constant speed u . If the distance between the trains is x , then what should be the minimum retardation of the train so as to avoid accident ?

(A) $\frac{(v+u)^2}{x}$

(B) $\frac{(v-u)^2}{x}$

(C) $\frac{(v+u)^2}{2x}$

(D) $\frac{(v-u)^2}{2x}$

Ans. (D)

Sol. The relative velocity of the train with respect to another train is

$$V_r = v - u; V^2 - V_r^2 = -2ax$$

$$0 - (v - u)^2 = -2ax; a = \frac{(v - u)^2}{2x}$$

- 40.** Two trains take 3 seconds to pass one another when going in opposite direction but only 2.5 seconds if the speed of one is increased by 50%. The time one would take to pass the other when they are going in same direction at original speeds is

(A) 12sec

(B) 10sec

(C) 18sec

(D) 15sec

Ans. (D)

Sol. Lengths of two trains are l_1 and l_2 and their speeds are v_1 and v_2 respectively.

$$\frac{l_1 + l_2}{v_1 + v_2} = 3 \text{ --- (1)} \quad \frac{l_1 + l_2}{v_1 + 1.5v_2} = 2.5 \text{ --- (2)} \quad \frac{(1)}{(2)}$$

$$\Rightarrow \frac{v_1 + 1.5v_2}{v_1 + v_2} = \frac{3}{2.5} = \frac{6}{5} \Rightarrow 5v_1 + 7.5v_2 = 6v_1 + 6v_2 \Rightarrow v_1 = 1.5v_2$$

$$\therefore \frac{l_1 + l_2}{2.5v_2} = 3 \Rightarrow l_1 + l_2 = 7.5v_2$$

$$t = \frac{l_1 + l_2}{v_1 - v_2} = \frac{7.5v_2}{1.5v_2 - v_2} = 15 \text{ sec}$$

- 41.** A body moves so that it follows the following relation $\frac{dv}{dt} = -v^2 + 2v - 1$ where v is speed in m/s and t is time in second. If at $t = 0$, $v = 0$ then choose the incorrect statement
- (A) terminal velocity is 1 m/s
 (B) the magnitude of initial acceleration is 1 m/s^2
 (C) instantaneous speed is $\frac{-1}{1+t}$
 (D) the speed is 1.5 m/s when acceleration is one fourth of its initial value

Ans. (C)

Sol. $\frac{dv}{dt} = -v^2 + 2v - 1$

(i) at terminal velocity (v_t)

$$a = 0$$

$$\Rightarrow 0 = -v^2 + 2v - 1 = (v - 1)^2$$

$$v = 1 \text{ ms}^{-1}$$

(ii) $\frac{dv}{dt} = \text{acceleration}$

$$a = -v^2 + 2v - 1$$

$$\text{at } t = 0, v = 0$$

$$\therefore a = -1 \text{ ms}^{-2}$$

$\therefore$ magnitude of acceleration

$$a = 1 \text{ ms}^{-2}$$

(iii) $\frac{dv}{dt} = -v^2 + 2v - 1 = -(v - 1)^2$

$$\Rightarrow \frac{dv}{(v - 1)^2} = -dt$$

$$\int_0^v \frac{dv}{(v - 1)^2} = -\int_0^t dt$$

$$\left(-\frac{1}{v - 1} \right)_0^v = (-t)$$

$$\Rightarrow \left(\frac{1}{v - 1} - \frac{1}{-1} \right) = t$$

$$\frac{v}{(v - 1)} = t$$

$$\Rightarrow 1 - \frac{1}{v} = \frac{1}{t}$$

$$\left(v = \frac{t}{t - 1} \right)$$

$\therefore$ (iii) is wrong

Note in option (C) make

$$-1 \Rightarrow +1$$

$$\text{i.e., } \frac{1}{(1+t)}$$

$$a_{\text{initial}} = -1 \text{ ms}^{-2}$$

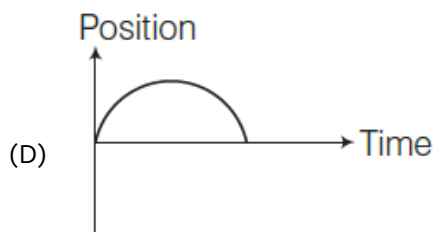
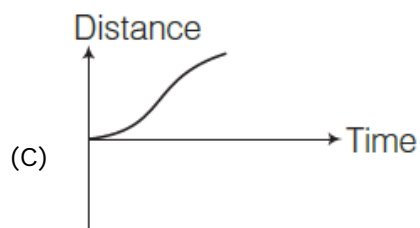
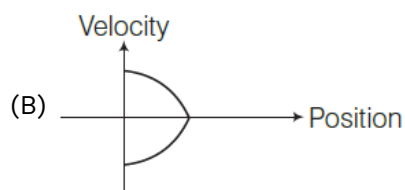
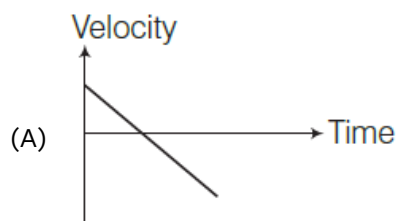
$$\therefore -\frac{1}{4} = -v^2 + 2v - 1$$

$$-\frac{1}{4} = -(v-1)^2$$

$$(v-1) = \frac{1}{2}$$

$$v = 1.5 \text{ ms}^{-1}$$

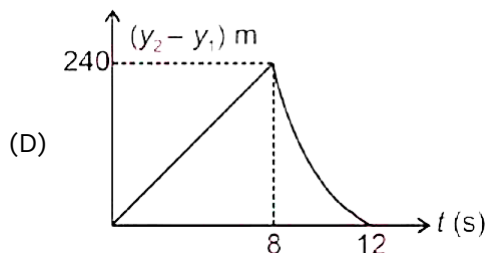
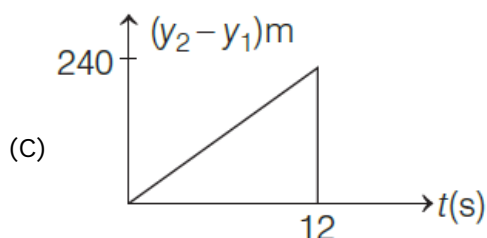
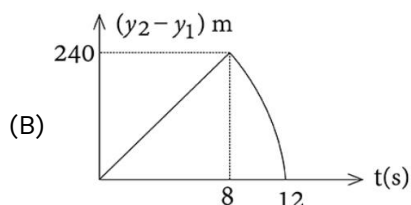
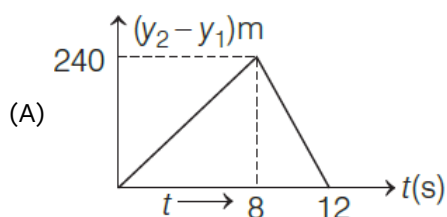
- 42.** All the graphs below are intended to represent the same motion. One of them does it incorrectly. Pick it up.



Ans. (C)

Sol. Graph A, B and D represent uniform retarded motion while graph C represents uniform accelerated motion initially.

43. Two stones are thrown up simultaneously from the edge of a cliff 240 m high with initial speed of 10 m/s and 40 m/s, respectively. Which of the following graph best represents the time variation of relative position of the second stone with respect to the first? (Assume stones do not rebound after hitting the ground and neglect air resistance, take $g = 10 \text{ m/s}^2$)



Ans. (B)

Sol. For the first ball,

$$-240 = 10t - \frac{1}{2}gt^2$$

$$\therefore 5t^2 - 10t - 240$$

$$\therefore t^2 - 2t - 48 = 0$$

$$\Rightarrow t = 8, -6$$

The 1st particle will reach the ground in 8 secs.

Upto 8 secs, the relative velocity between the particles is 30m/sec and the relative acceleration is zero..

For the 2nd particle,

$$-240 = 40t - 5t^2 \Rightarrow t^2 - 8t - 48 = 0$$

$$t = 12 \text{ secs}$$

The second particle will strike the ground in 12 secs.

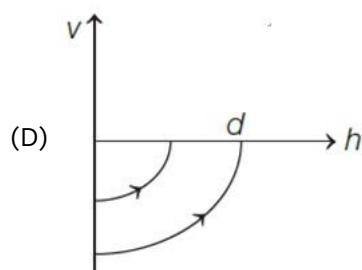
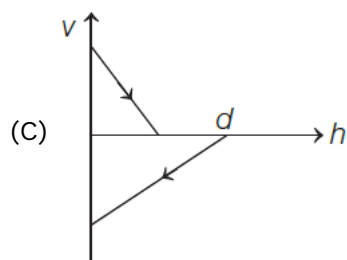
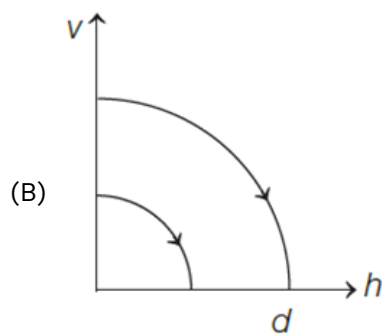
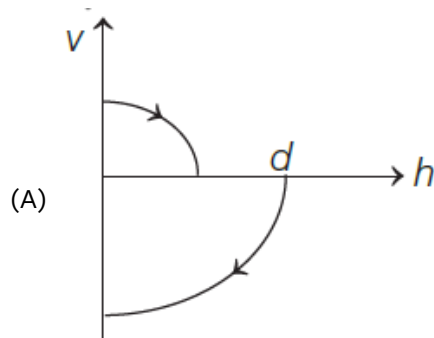
0 ----- 8 sec, $v_{\text{rel}} = \text{constant}$

8 ----- 12sec, $a_{rel} \neq 0$, 1st stone strikes the ground and comes to rest

$$\therefore a_{rel} = g(\downarrow) \text{ i.e. } a_{rel} = -ve$$

$\therefore y_{rel}$ graph will be downward parabola

- 44.** A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistance, its velocity v varies with height h above the ground as



Ans. (A)

Sol. Initial height = $+d$, $v = 0$, as time passes ' h ' decreases and v increases in downward direction ($-ve$). After striking the ground ' v ' reverses (changes sign) and also reduces (because it only rebounds to height $d/2$). Also ' v ' at any instant varies non-linearly with ' h '

$$v^2 = 2g(d - h)$$

45. Let $|A_1| = 3$, $|A_2| = 5$ and $|A_1 + A_2| = 5$. The value of $(2A_1 + 3A_2) \cdot (3A_1 - 2A_2)$ is
- (A) -106.5
(B) -112.5
(C) -99.5
(D) -118.5

Ans. (D)

Sol. $|\vec{A}_1| = 3 \quad |\vec{A}_2| = 5 \quad |\vec{A}_1 + \vec{A}_2| = 5$

$$|\vec{A}_1 + \vec{A}_2| = \sqrt{|\vec{A}_1|^2 + |\vec{A}_2|^2 + 2|\vec{A}_1||\vec{A}_2|\cos\theta}$$

$$5 = \sqrt{9 + 25 + 2 \times 3 \times 5 \cos\theta}$$

$$\cos\theta = -\frac{9}{2 \times 3 \times 5} = -\frac{3}{10}$$

$$(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$$

$$= 6|\vec{A}_1|^2 + 9\vec{A}_1 \cdot \vec{A}_2 - 4\vec{A}_1\vec{A}_2 - 6|\vec{A}_2|^2$$

$$54 + 5 \times 3 \times 5 \left(-\frac{3}{10}\right) - 6 \times 25$$

$$= 54 - 150 - \frac{45}{2} = -118.5$$

46. The position vector of a particle is given as $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to :
- (A) 1 sec
(B) 2 sec
(C) 1.5 sec
(D) Not possible

Ans. (A)

Sol. $\vec{V} \cdot \vec{a} = 0$ Expression of $\vec{v}$ & $\vec{a}$

47. Two particles having position vectors $\vec{r}_1 = (3\hat{i} + 5\hat{j})$ meters and $\vec{r}_2 = (-5\hat{i} - 3\hat{j})$ meters are moving with velocities $\vec{v}_1 = (4\hat{i} + 3\hat{j})$ m/s and $\vec{v}_2 = (\alpha\hat{i} + 7\hat{j})$ m/s. If they collide after 2s, the value of α is

:-

- (A) 2
(B) 4
(C) 6
(D) 8

Ans. (D)

Sol. $\vec{v}_1 t + \vec{r}_1 = \vec{v}_2 t + \vec{r}_2$ is condition for collision

- 48.** A particle is projected with an initial velocity of 200m/s in a direction making an angle of 30° with the vertical. The horizontal distance covered by the particle in 3s is
- (A) 300 m
(B) 150 m
(C) 175 m
(D) 125m

Ans. (A)

Sol. $x = u \cos \theta t = 200 \times \cos 60^\circ \times 3 = 600 \times \frac{1}{2} = 300m$

- 49.** From the top of a tower 39.2 m high, a ball is thrown horizontally. If line joining the point of projection to the point where it hits the ground makes an angle of 45° with the horizontal, then the initial velocity of the ball is:
- (A) 9.8 ms^{-1}
(B) 4.9 ms^{-1}
(C) $9.8\sqrt{2} \text{ ms}^{-1}$
(D) 19.6 ms^{-1}

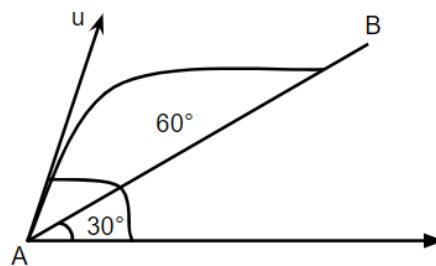
Ans. (C)

Sol. $\tan \theta = \frac{H}{R} \Rightarrow \theta = 45^\circ$

$$H = V \times \sqrt{\frac{2H}{g}}$$

$$V = \sqrt{\frac{gH}{2}} = \sqrt{\frac{g \times 4g}{2}} = g\sqrt{2}$$

- 50.** Time taken by the projectile to reach from A to B is t , then the distance AB is equal to :



- (A) $\frac{ut}{\sqrt{3}}$
(B) $\frac{\sqrt{3} ut}{2}$
(C) $\sqrt{3} ut$
(D) $2ut$

Ans. (A)

Sol. From the diagram $AC = u \cos 60t = \frac{ut}{2}$

$$AB = \frac{AC}{\cos 30} = \frac{ut}{2 \left(\frac{\sqrt{3}}{2} \right)} = \frac{ut}{\sqrt{3}}$$

SECTION-II

- 51.** A football is thrown with a velocity of 10 ms^{-1} at an angle of 30° above the horizontal. What will be the time of flight, in seconds? ($g = 10 \text{ ms}^{-2}$)

Ans. 1

Sol. $T = \frac{2u \sin \theta}{g} = \frac{2 \times 10 \times \sin 30^\circ}{10} = 1$

- 52.** The minimum speed (in ms^{-1}) with which a projectile must be thrown from origin at ground so that it is able to pass through a point P (30 m, 40 m) is γ then the value of $\frac{\gamma}{10}$ is ($g = 10 \text{ ms}^{-2}$)

Ans. 3

Sol. Let u and θ be the velocity and angle of projection respectively.

For the projectile to pass through P(x, y)

$$y = x \tan \theta - \frac{gx^2}{2u^2} (1 + \tan^2 \theta)$$

$$\Rightarrow gx^2 \tan^2 \theta - 2xu^2 \tan \theta + (gx^2 + 2yu^2) = 0$$

The projectile will pass through P(x, y) if this equation (quadratic in $\tan \theta$) gives some real value of θ , i.e., its discriminant ≥ 0 .

$$4x^2u^4 - 4gx^2(gx^2 + 2yu^2) \geq 0$$

$$u^4 - 2gyu^2 - g^2x^2 \geq 0$$

$$u^4 - 2gyu^2 + g^2y^2 \geq g^2x^2 + g^2y^2$$

$$(u^2 - gy)^2 \geq (x^2 + y^2)g^2$$

$$\Rightarrow u \geq \sqrt{gy + g\sqrt{x^2 + y^2}}$$

So, the minimum value of u is $u_{\text{MIN}} = \sqrt{gy + g\sqrt{x^2 + y^2}}$.

$$u_{\text{min}} = \sqrt{10 \times 40 + 10\sqrt{30^2 + 40^2}} = \sqrt{900}$$

$$u_{\text{min}} = 30 \text{ ms}^{-1}$$

$$\therefore \gamma = 30; \therefore \frac{\gamma}{10} = \frac{30}{10} = 3$$

- 53.** Trajectory of particle launched obliquely from the ground is given as $y = x - \frac{x^2}{8}$, where, x and y are in metre. For this projectile motion angle of projection is θ degrees, then $\frac{\theta}{15}$ is, if $g = 10 \text{ ms}^{-2}$

Ans. 3

Sol. Given $y = x \left(1 - \frac{x}{80} \right)$

Comparing with the standard form

$$y = x \tan \theta \left(1 - \frac{x}{R} \right), \Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

$$\frac{\theta}{15} = \frac{45}{15} = 3$$

- 54.** When two particles A and B are at point O, A is moving with a constant velocity 50 ms^{-1} , while B is not moving. But B possesses a constant acceleration of 10 ms^{-2} . After how much time they will be at a separation of 125 m?

Ans. 5

Sol. For particle A,

$$u_A = 50 \text{ ms}^{-1}$$

$$a_A = 0 \text{ ms}^{-2}$$



For particle B

$$u_B = 0$$

$$a_B = 10 \text{ ms}^{-2}$$

So, initial velocity of A w.r.t. B

$$u_{AB} = u_A - u_B = 50 \text{ ms}^{-1}$$

and acceleration of A w.r.t. B

$$a_{AB} = a_A - a_B = -10 \text{ ms}^{-2}$$

The distance between A and B after time t is given by

$$s_{AB} = u_{AB}t + \frac{1}{2}a_{AB}t^2$$

$$\Rightarrow 125 = 50t - 5t^2$$

$$\Rightarrow t^2 - 10t + 25 = 0$$

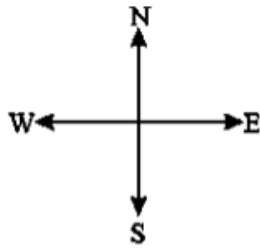
$$\Rightarrow (t - 5)^2 = 0$$

$$\Rightarrow t = 5\text{s}$$

- 55.** Car A has an acceleration of 2 ms^{-2} due east and car B, 4 ms^{-2} due north. What is the acceleration of car B with respect to car A? (take $\sqrt{5}=2.5$)

Ans. 5

Sol. This is a two - dimensional motion.



Let North and East directions by positive y - axis and x - axis respectively.

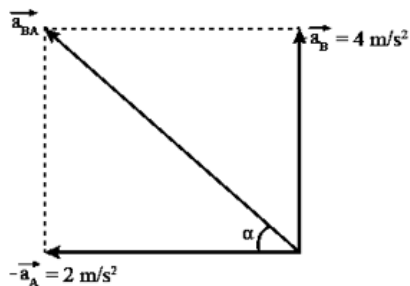
Therefore, $\vec{a}_{BA}$ = acceleration of car B with respect to car A

$$= \vec{a}_B - \vec{a}_A$$

Here, $\vec{a}_B$ = acceleration of B = $4\hat{j} \text{ m/s}^2$

And $\vec{a}_A$ = acceleration of A = $2\hat{i} \text{ m/s}^2$

$$\therefore \vec{a}_{BA} = (4\hat{j} - 2\hat{i}) \text{ m/s}^2$$



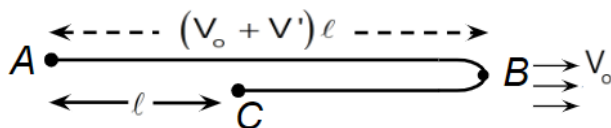
$$|\vec{a}_{BA}| = \sqrt{(4)^2 + (2)^2} = 2\sqrt{5} \text{ m/s}^2$$

$$\therefore \sqrt{5} = 2.5 \Rightarrow 2\sqrt{5} = 2 \times 2.5 = 5$$

- 56.** A motorboat going downstream overcame a raft at point A. One hour later it turned back and met the raft again at a distance 6 km from point A. Find the river velocity (in kmph).

Ans. 3

Sol. Considering the given figure below.



Let v_0 be the stream velocity and v' the velocity of motorboat with respect to water. The motorboat reached point B while going downstream with velocity $(v_0 + v')$ and then returned with velocity $(v' - v_0)$ and passed the raft at point C. Let t be the time for the raft (which flows with stream with velocity v_0) to move from point A to C, during which the motorboat moves from A to B and then from B to C. Therefore

$$\frac{l}{v_0} = \tau + \frac{(v_0 + v')\tau - l}{(v' - v_0)}$$

On solving, we get $v_0 = \frac{l}{2\tau} = 3 \text{ km/hr}$.

- 57.** Snow is falling vertically at a constant speed of 8 ms^{-1} . At angle β (in degree) from the vertical do the snow flakes appear to be falling as viewed by the driver of a car travelling on a straight, level road with a speed of 50 kmh^{-1} . Then the value of $\beta/10$ is. (nearly)

Ans. 6

Sol. Let V_v be the vertical velocity of snowflakes and V_c be the velocity of the car. The resultant velocity is V_{relative}

$$\theta = \tan^{-1} \left(\frac{V_c}{V_v} \right)$$

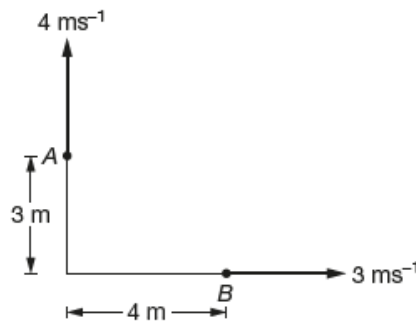
$$= \tan^{-1} \left(\frac{50 \times 10^3}{3600 \times 8} \right)$$

$$= \tan^{-1}(1.736)$$

$$\theta = 60^\circ$$

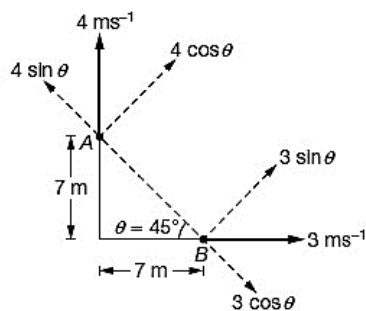
$$\therefore \beta = 60; \therefore \frac{\beta}{10} = \frac{60}{10} = 6$$

- 58.** Particles A and B are moving as shown in the figure at $t = 0$, their relative velocity of separation at $t = 1 \text{ s}$ is $\frac{\alpha}{\sqrt{\beta}}$ then $\alpha - \beta$ is



Ans. 5

Sol.



V_{sep} = relative velocity along line AB

$$\Rightarrow V_{\text{sep}} = 3 \cos \theta + 4 \sin \theta$$

$$\Rightarrow V_{\text{sep}} = 3 \left(\frac{1}{\sqrt{2}} \right) + 4 \left(\frac{1}{\sqrt{2}} \right) = \frac{7}{\sqrt{2}} \text{ ms}^{-1}$$

$$\therefore \frac{\alpha}{\sqrt{\beta}} = \frac{7}{\sqrt{2}}$$

$$\therefore \alpha - \beta = 7 - 2 = 5$$

59. If angular displacement of a particle is given by $\theta = a - bt + ct^2$, then find its angular velocity (rad/sec) at $t = 2\text{sec}$ if $b = 2$, $c = 1$.

Ans. 2

Sol. $\omega = \frac{d\theta}{dt} = -b + 2ct$
 $= -2 + 2 \times 1 \times 2 = 2$

60. A particle is moving with constant speed in a circular path. The ratio of average velocity to its instantaneous velocity when the particle describes an angle $\theta = \frac{\pi}{2}$ is $\frac{p\sqrt{q}}{\pi}$, then $\frac{p}{q}$ is

Ans. 1

Sol. Time taken to describe angle θ

$$t = \frac{\theta}{\omega} = \frac{\theta R}{v} = \frac{\pi R}{2v}$$

$$v_{av} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\sqrt{2}R}{\pi R / 2v} = \frac{2\sqrt{2}}{\pi} v$$

Instantaneous velocity = v

The ratio of average velocity to its instantaneous velocity is

$$\frac{v_{av}}{v_{ins}} = \frac{2\sqrt{2}}{\pi}$$

$$\therefore \frac{p\sqrt{q}}{\pi} = \frac{2\sqrt{2}}{\pi}$$

$$p = 2, q = 2 \Rightarrow \frac{p}{q} = \frac{2}{2} = 1$$

PART-C: CHEMISTRY

SECTION-I

61. The number of atoms present in 16 g of oxygen gas is:

- (A) $6.02 \times 10^{11.5}$
 (B) 3.01×10^{23}
 (C) $3.01 \times 10^{11.5}$
 (D) 6.02×10^{23}

Ans. (D)

Sol. Moles of $O_2 = 16/32 = 0.5$

Moles of atoms = $0.5 \times 2 = 1$

Number of atoms = $1 \times N_A = 6.02 \times 10^{23}$

62. A signature, written in carbon pencil weights 1 mg. What is the number of carbon atoms present in the signature?

- (A) 5.02×10^{23}
 (B) 5.02×10^{20}
 (C) 6.02×10^{20}
 (D) 0.502×10^{20}

Ans. (D)

Sol. 12 g of carbon contain 6.02×10^{23} carbon atoms 1mg (10^{-3} g) carbon contain x

$$x = \frac{10^{-3} \times 6.02 \times 10^{23}}{12} = .5018 \times 10^{20} \text{ atom}$$

63. Cortisone is a molecular substance containing 21 atoms of carbon per molecule. The molecular weight of cortisone is 360.4. what is the percentage of carbon in cortisone?

- (A) 59.9%
- (B) 75%
- (C) 69.92%
- (D) None of these

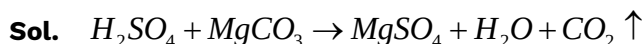
Ans. (C)

Sol. $\%C = \frac{12 \times 21}{360.4} \times 100 = 69.92\%$

64. The weight of sulphuric acid needed for dissolving 3 g magnesium carbonate is:

- (A) 3.5 g
- (B) 7.0 g
- (C) 1.7 g
- (D) 17.0 g

Ans. (A)



1mole 1mole

98g $\rightarrow$ 84g

x $\rightarrow$ 3g

$$x = \frac{3 \times 98}{84} = 3.5g$$

65. 5.85 g of NaCl dissolved in H_2O and solution is made upto 500 mL. The molarity is:

- (A) 0.1
- (B) 0.2
- (C) 1.0
- (D) 0.117

Ans. (B)

Sol. $M = \frac{5.85}{58.5} \times \frac{1000}{500} = 0.2M$

66. The mass of 112 cm³ of CH_4 gas at STP is

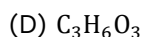
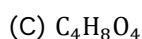
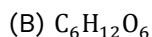
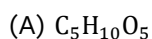
- (A) 0.16 g
- (B) 0.8 g
- (C) 0.08 g
- (D) 1.6 g

Ans. (C)

Sol. mole of $CH_4 = \frac{112}{22400} = 0.5 \times 10^{-2}$

Mass = $0.5 \times 10^{-2} \times 16 = 0.08g$

67. 0.1 mole of a carbohydrate with empirical formula CH_2O contains 1 g of hydrogen. What is its molecular formula?



Ans. (A)

Sol. 0.1 mole carbohydrate $\rightarrow$ 1g Hydrogen

1 mole $\rightarrow$ 10 g Hydrogen

Means 10-Hydrogen atoms

Molecular formula = $C_5H_{10}O_5$

68. If 8 litre of Oxygen was left reacted, when 4 litre of H_2 reacted with y lit. of O_2 at STP. find y ?

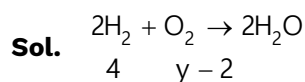
(A) 8 lit

(B) 12 lit

(C) 10 lit

(D) 6 lit

Ans. (C)



$y - 2 = 8$

$y = 10$ lit

69. For the reaction, $A + 2B \rightarrow C$, 5 moles of A and 8 moles of B will produce:

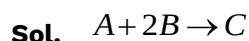
(A) 5 moles of C

(B) 4 moles of C

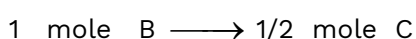
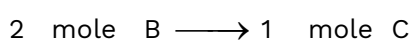
(C) 8 moles of C

(D) 13 moles of C

Ans. (B)

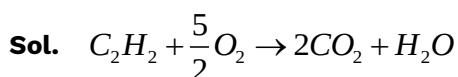


Limiting reagent is "B"



- 70.** Air contains 20% O_2 by volume. How much volume of air will be required for complete combustion of 100 cc of acetylene?
- (A) 500 cc
(B) 1064 cc
(C) 212.8 cc
(D) 1250 cc

Ans. (D)



$$22.4 \rightarrow \left(\frac{5}{2} \times 22.4 \right) \text{ litre}$$

$$0.1 \text{ litre} \rightarrow \left(x \times \frac{20}{100} \right)$$

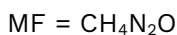
Since 20% O_2 is present in air

$$x = 1.25L \quad (\text{or}) \quad 1250 \text{ cc}$$

- 71.** An organic compound contains 20.0% C, 6.66% H, 47.33% N and the rest was oxygen. Its molar mass is 60 g mol^{-1} the molecular formula of the compound is
- (A) CH_4N_2O
(B) $C_2H_4NO_2$
(C) CH_3N_2O
(D) $CH_4N_2O_2$

Ans. (A)

Sol. Empirical formula concept



- 72.** Dissolving 120 g of urea (mol.wt.60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is:
- (A) 1.78 M
(B) 2.00 M
(C) 2.05 M
(D) 2.22 M

Ans. (C)

Sol. $\text{molality (m)} = \frac{120}{60} \times \frac{1000}{1000} = 2$

d = density, m = molality, M = Molarity

Use formula,
$$m = \frac{M \times 1000}{(d \times 1000) - (M \times GMW \text{ of Solute})}$$

- 73.** The mass of nitrogen per gram hydrogen in the compound hydrazine is exactly one and half times the mass of nitrogen in the compound ammonia. The fact illustrates the
- (A) Law of conservation of mass
(B) Multiple valency of nitrogen

- (C) Law of multiple proportions
(D) Law of definite proportions

Ans. (C)

Sol. Law of multiple proportions

74. 20% w/w aq. solution of NaOH having density 1.5 g/ml find molarity (M)?

- (A) 2.5
(B) 7.5
(C) 5
(D) 25

Ans. (B)

Sol. $M = \frac{10 \times d}{M \cdot wt} = \frac{10 \times 20 \times 1.5}{40} = 7.5$

75. The number of atoms in 4.25 g of NH_3 is approximately

- (A) 6×10^{23}
(B) 2×10^{23}
(C) 1.5×10^{23}
(D) 1×10^{23}

Ans. (A)

Sol. 17 g of NH_3 ----- $4 \times N_A$ of atoms
4.25g ----- x

$$x = \frac{4.25 \times 4 \times N_A}{17} = 6.37 \times 10^{23}$$

76. 100 g of $CaCO_3$ is treated with 1 L of 1 M HCl. What would be the weight of CO_2 liberated after the completion of the reaction?

- (A) 55 g
(B) 11 g
(C) 22 g
(D) 33 g

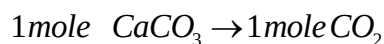
Ans. (C)

Sol. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2 \uparrow$

HCl is the limiting reagent

$$= M_2 V_2 = 50 \times 0.1 = 5 \text{ Equivalent (or) 1 mole of HCl will react with 0.5g } CaCO_3$$

$\Rightarrow$ From equation

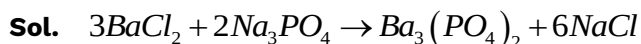


$\therefore$ Weight of $CO_2 = 22g$

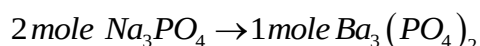
77. If 0.5 mole of $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4 the maximum number of mole of $Ba_3(PO_4)_2$ that can be formed is:

- (A) 0.7
(B) 0.5
(C) 0.30
(D) 0.1

Ans. (D)



0.5 mole 0.2 mole (L.R)



$0.2 \text{ mole} \rightarrow 0.1 \text{ mole}$

78. The number of hydrogen atoms present in 25.6 g of sucrose($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) which has a molar mass of 342 g is

(A) 22×10^{23}

(B) 9.91×10^{23}

(C) 11×10^{23}

(D) 44×10^{23}

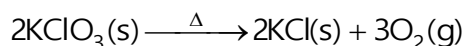
Ans. (B)

Sol. 342g (1 mole) sucrose contain ----- $12 \times N_A$ of Hydrogen

25.6 g sucrose ----- x

$$x = \frac{25.6 \times 22 \times N_A}{342} = 9.91 \times 10^{23}$$

79. For given balanced Equation



volume of O_2 released at STP (1 mole gas = 22.4 lit) when $\frac{3}{2}$ moles of KClO_3 heated is?

(A) 50.4 lit

(B) 33.6 lit

(C) 22.4 lit

(D) 44.8 lit

Ans. (A)

Sol. 2 mole $\text{KClO}_3 = 3$ mole O_2

1.5 mole $\text{KClO}_3 = 2.25$ mole O_2

$V_{\text{O}_2} = 2.25 \times 22.4 = 50.4$ lit.

80. There are two isotopes of an element with average atomic mass z . Heavier one has atomic mass $z + 2$ and lighter one has $z - 1$, the abundance of lighter one is

(A) 66.6%

(B) 69.7%

(C) 6.67%

(D) 33.3%

Ans. (A)

Sol. Let % abundancy of Heavier Isotope = a

% abundancy of lighter Isotope = (100 - a)

Average atomic mass = z

$\therefore$ Avg. atomic mass

$$= \sum \frac{(\%abundancy \times atomic\ mass\ of\ Isotope)}{100}$$

By using above formula

Percentage abundancy of lighter isotope = $(100 - a) = 66.67\%$

SECTION-II

81. If one mole of H_2SO_4 reacts with excess of $NaOH$. How many moles of H_2O will be formed _____.

Ans. 2

Sol. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

82. An element A (at.wt = 75) and B (at.wt = 25) combine to form a compound. The compound contains 75% of A by weight. Total number of atoms present in the simple formula of molecule ____.

Ans. 2

Sol. Empirical formula Calculation

83. 1g $NaOH$ reacted with 25 ml of 0.75 M HCl solution mass of $NaOH$ left unreacted in mg is x. Then

find $\frac{2x}{100}$?

Ans. 5

Sol. $NaOH + HCl \longrightarrow NaCl + H_2O$

$$\frac{1000}{40}$$

$$25 \times 0.75$$

$$= 25 \text{ m mol} \quad \Rightarrow \frac{75}{4} \text{ m mol}$$

HCl is LR

So m mol of $NaOH$ left unreacted is $\frac{25}{4}$

$$\text{wt. of } NaOH \text{ unreacted} \Rightarrow \frac{25}{4} \times 40 = 250 \text{ mg} = x$$

$$\frac{2 \times 250}{100} = 5$$

84. The gas is found to have formula $(CO)_x$. Its vapour density is 70, the value of "x" is

Ans. 5

Sol. mol. wt = $2 \times V.D$

$$= 2 \times 70 = 140$$

$$\text{For } (CO)_x, x(12 + 16) = 140$$

$$28x = 140$$

$$\therefore x = \frac{140}{28} = 5$$

85. 4 gram atom of silver (Ag) contains $P \times 10^2$ grams of silver metal. The value of P to the nearest single digit integer ____.

Ans. 4

Sol. 1 gram atom of silver means 108 g or 1 mole

$$4 \text{ grams atom means } 4 \times 108 = 432 \text{ g} = 4.32 \times 10^2 \text{ g}$$

86. If 20% nitrogen is present in a compound, its minimum molecular weight is obtained as "P". Find the value of P/10?

Ans. 7

$$\text{Sol. } 20\% = \frac{14 \times 1}{x} \times 100 \Rightarrow x = \frac{1400}{20} = 70$$

87. Mole fraction of I_2 in benzene is 0.2 find molality?

(M. wt. of I_2 = 254 g/mole, C_6H_6 = 78 g/mole) (nearest integer)

Ans. 3

$$\text{Sol. } m = \frac{0.2 \times 1000}{0.8 \times 78} = 3.2 \approx 3$$

88. Total moles of atoms present in 128g of SO_2 gas at STP is (are) ____.

Ans. 6

Sol. 64 g of SO_2 ----- 3 mole of atoms

128 g ----- 6 mole

89. 0.0833 mole of Carbohydrate of Empirical formula CH_2O contain 1g of Hydrogen. Total number of Hydrogen atoms present in molecular formula of the compound is 'x'. Then the value of x/4 is ____.

Ans. 3

Sol. 0.0833 mole carbohydrate ----- 1g hydrogen

1 mole contain ----- $1/0.0833 = 100/8.33 = 12$

So, the molecular formula have 12-Hydrogen atoms

90. 4.9% $\frac{W}{V}$ H_2SO_4 solution. The Molarity M of the solution is x. Then find the value of 10x.

Ans. 5

Sol. 4.9 g H_2SO_4 in 100 ml solution.

$$M = \frac{4.9}{98} \times \frac{1000}{100} = \frac{1}{2} = 0.5M$$